

Memory resistor

Memristor technology could lead to computer systems that can remember and associate patterns in a way similar to humans

VIS 4D

Ethereal Technologies' Volumetric Imaging System or VIS4D can be used by gamers to visualise 3D images in real time without glasses

Tomorrow's Tech

Smartbooks?

At a recent round-table in Mumbai, NVIDIA's CEO, Jen-Hsun Huang was in good spirits, reflecting that balloons in the movie *Up* wouldn't brush a certain



In case you're lost, use Layar

way, if it weren't for NVIDIA's contribution to graphics technology. None of the journalists present were willing to question that statement.

On that particular day, Intel had just agreed on an out-of-court settlement, to a lawsuit filed by AMD, for a sum of \$1.25 billion. Jen-Hsun opined that when compared to the profits Intel made during the period of the lawsuit, it was the equivalent of paying for a parking ticket.

On display at the meeting were a wide variety of Tegra-based devices – e-book readers, smartphones, netbooks, and a large touch tablet. "The beauty of low-power design is that the form factor can be exquisite. The hi-def movie was streaming off a tiny Mobinova netbook. It can play hi-def movies for 12 hours, and gives week long battery life otherwise", Jen-Hsun said.

The netbook has been attributed as an Asian innovation. Starting from the EEE PC, it has provided a low-power and low-cost computing option to a large section of society, and is currently available for \$300 to \$600. This platform has been dominated by Intel so far, who have also broadly defined the specifications and limitations of a device of this form

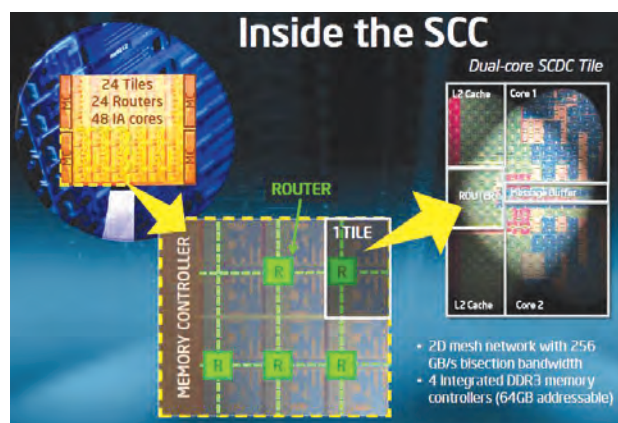
factor, to prevent it from overlapping with the laptop market. Not everyone is happy with this particular arrangement.

Unlike Intel, ARM puts no limitations on the size or form of the products that are made using it. The Tegra, five years in the making is based on the ARM11 core. "We wanted to create a computer that is as good at 1/50th the power", said Jen-Hsun Huang.

Crystal gazing into the future of handhelds/netbooks, or at least the processors that power them is not that difficult. The ARM 11 SOC (system on a chip), which powers a majority of Android devices and the original Apple iPhone, was first announced in 2003. In Anand Shimpi's column titled Core Values, he explains that the Cortex A8, which debuted in 2005, took four years to be incorporated into a product, first on the iPhone 3GS

and Palm Pre. It takes time, because ARM does not manufacture processors like AMD or Intel, but licenses its design to players like Texas Instruments, Samsung, Qualcomm, and NVIDIA, who make their own customised SoCs.

We can safely predict a new product lineup called smartbooks this year, with emphasis on touch-based computing,



Processor cores are increasing by the day

and as a communications tool with data stored on a cloud. At less than Rs. 10,000, these will be cheaper than Intel's Atom netbooks, or in a different usage model, likely to be bundled free with a wireless internet telecom service provider.

Tegra 2 is expected this year. Over a five year lifecycle, we are also very likely to see implementations of the Cortex A9, which will be produced by AMD's spinoff GlobalFoundries. These will power not just touch-based smart phones and tablets, but netbooks and laptops as well. While these devices will offer an entirely new user experience compared to Windows PCs or Apple Macs, they will be able to run Google's Chrome OS, and store data on a cloud. They will boot in seconds, always be connected – either via inbuilt 3G or wireless connectivity – and have a much longer battery life. And you don't have to every worry about backing up your data.

Interfaces

In a TED talk about simplicity and design. NY Times columnist David Pogue mentions how Palm Inc. used to have an employee whose job designation was that of a "tap counter" – to ensure that all their applications achieve their desired outcome in three taps. Of late, everyone seems to have embraced this aesthetic. However, two companies, Swype and First Else, have come up with new interface designs this year that seem like the next evolutionary leap. In terms of speed and elegance, they leave the iPhone, the holy grail of UI design in the dust.

The First Else phone has been made by an Israeli design house and draws comparisons to sci-fi user interfaces seen in *Minority Report*, *Fifth Element*, and *Terminator*. The device does away with icons, and tries to adapt to you, instead of the other way around. One just slides the finger, letting the menu options fan out. The highlighted item gets selected once you lift your finger. The

radial fan-like menu spreads out options based on thumb position, contacts are sorted by call frequency, First Else also claims to provide cross platform search, and backs up all your contacts and information on the cloud.